Max Dubowski

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Education

Heriot-Watt University — Edinburgh, UK
MEng (Hons) Aerospace Engineering — *Sep 2024 – May 2028*

- **Academic Honors:** Awarded the **Deputy Principal's Award** for straight Grade A performance across all Year 1 engineering modules.
- **Core Coursework:** Flight Mechanics, Aerodynamics, Fluid Dynamics, Finite Element Analysis (FEA), Mathematics for Engineers.

Heriot-Watt University Malaysia — Putrajaya, Malaysia
Study Abroad Exchange Semester – Aerospace Engineering — *Sep 2025 – Dec 2025*

- Partnered with peers from diverse international backgrounds, refining intercultural communication, active listening, and teamwork essential for global engineering projects.
- Cultivated personal autonomy and resilience 6,000 miles from home, learning to independent travel across Southeast Asia (Singapore, Hong Kong, Thailand).

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, MATLAB, Swift, HTML/CSS
Full-Stack & Backend: Next.js, React, Node.js, FastAPI, REST APIs, Firebase, Leaflet, Tailwind CSS, Vercel
Engineering & Simulation: ANSYS Fluent (CFD), Abaqus FEA, SpaceClaim, Autodesk Inventor, 3D Printing (FDM)
Developer Tools: Git/GitHub, Docker, Linux/Bash, Google AI Studio / Gemini API, VS Code
Spoken Languages: English (Native), Polish (Native)

Experience

Packsync (packsync.co.uk) — Edinburgh, UK
Technical Co-Founder & Software Engineer — *Sep 2026 – Present*

- Co-founded and engineered a B2B SaaS platform automating packaging compliance for UK beverage producers.
- Architected responsive web application using **Next.js 16**, **TypeScript**, and **Tailwind CSS**, deployed on **Vercel**.
- Built real-time compliance calculation engines parsing complex producer material streams and fee schedules.
- Integrated automated transactional client notification pipelines utilizing the **Resend API**.

Private Residential Portfolio — Edinburgh, UK
Assistant Property Manager (Self-employed) — *Feb 2025 – Jan 2026*

- Managed tenant operations, check-in logistics, and turnover inspections across high-occupancy residential properties.
- Conducted hands-on technical fault diagnostics across plumbing, electrical, and structural systems, coordinating trade contractors.

Technical Projects

BACKED – AI Prediction Market Engine | *Python, Gemini API (Gemma 4), WebSockets, Multimodal* — Jul 2026

- Placed 2nd in the google hackathon with a architected multi-threaded Python backend engine integrating **Gemma 4** via the Gemini API to run a live social prediction market on study commitments at **Build with Gemma: GDGoC Aberdeen**.
- Configured dual reasoning pipelines: calibrated probability pricing over persistent learner history JSON models (*thinking: high*) and low-latency mid-session margin call nudges.
- Built systems resilience with input-hash caching, automated exponential backoff for rate limits, defensive JSON parsing, and multi-threaded request decoupling.

Friendly – Real-Time Campus Coordination App | *Next.js 16, TypeScript, React 19, Firebase, Leaflet* — 2026

- Built a real-time social coordination platform for university cohorts at **RGU Hack 2026**.
- Engineered a sweep-line interval algorithm to process group schedule blocks and automatically surface optimal shared study windows without manual polling.
- Integrated an interactive campus map with Leaflet and React-Leaflet featuring real-time location sharing and 5 distinct study status indicators.
- Implemented Firebase Anonymous Authentication for zero-friction onboarding alongside Firestore real-time listeners.

Missile Trajectory & Guidance Simulator | *Python, NumPy, Matplotlib* — 2026

- Developed a dynamic physics simulator predicting missile flight paths via 3-DOF equations of motion using Runge-Kutta integration.
- Integrated aerodynamic drag curves, variable mass burn, and proportional navigation guidance algorithms to intercept moving targets.

Structural Finite Element Analysis Tool | *C++* — 2025

- Programmed a standalone solver computing shear force, bending moments, and deflection curves for indeterminate beam structures.
- Implemented matrix boundary-condition solvers to validate theoretical structural mechanics against numerical FEA results.